MARKED z-PRIMES ATTACHED TO THE NONTRIVIAL ZEROS OF THE RIEMANN ZETA FUNCTION: EXACT QUANTIZATION, PRIME-POWER RESONANCES, ASYMPTOTIC BOUNDS, AND A THEORETICAL LOCALIZATION LATTICE

ABSTRACT. Let p_k be the k-th ordinary prime and let $\gamma_k > 0$ be the k-th positive ordinate of a nontrivial zero of the Riemann zeta function, counted with multiplicity. We define the marked z-prime attached to γ_k by

$$n_k = \left\lceil \frac{\gamma_k \log p_k}{\pi} \right\rceil, \qquad z_k = \exp\left(\frac{\pi n_k}{\gamma_k}\right), \qquad \mathfrak{z}_k = (n_k, z_k). \tag{0.1}$$

The integer mark is essential: the pair (n_k, z_k) recovers the ordinate exactly through $\gamma_k = \pi n_k / \log z_k$, whereas z_k alone retains only a phase class. The construction is local, canonical, and need not be monotone in the second coordinate. We prove exact quantization identities, a bijection between the positive zero ordinates and the marked system, sharp one-cell bounds, the asymptotics $n_k \sim 2k$, $z_k \sim p_k$, and

$$z_k - p_k = \left(\frac{\varepsilon_k}{2} + o(1)\right)(\log k)^2, \qquad 0 \le \varepsilon_k < 1. \tag{0.2}$$

Consequently the upper envelope satisfies $\limsup(z_k - p_k)/(\log k)^2 \le 1/2$, and the marked counting function has the same first-order density as the ordinary primes.

The logarithmic coordinate $\log z_k$ is the first node above $\log p_k$ in the variable lattice $(\pi/\gamma_k)\mathbb{Z}$. This produces exact formulae for the detuning of the Mangoldt frequencies $r \log p_k$ from signed resonant nodes. In particular,

$$\mathrm{e}^{\mathrm{i}\gamma_k \log p_k^r} = (-1)^{r n_k} \mathrm{e}^{-\mathrm{i}\pi r \varepsilon_k}, \tag{0.3}$$

and the distance to the nearest signed resonance is $\pi \|r\varepsilon_k\|_{\mathbb{R}/\mathbb{Z}}$. Dirichlet approximation therefore produces arbitrarily accurate resonant prime powers, while the factor $p_k^{-r/2} \log p_k$ from the explicit formula suppresses high powers. Finally, we formulate a bivariate marked Dirichlet transform and an inverse lattice program in which the pairs (n, z) serve as exact coordinates for zero ordinates. All inverse-selection statements beyond the exact transform identities are isolated as open analytic problems.

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1. Introduction

The nontrivial zeros of the Riemann zeta function are written

$$\rho_k = \beta_k + \mathrm{i}\gamma_k, \qquad 0 < \gamma_1 \le \gamma_2 \le \cdots, \tag{1.1}$$

where ordinates are counted with multiplicity. The classical Riemann–von Mangoldt formula controls their counting function, while the prime number theorem controls the ordered primes p_k; standard references are [6, 5]. The present construction combines the k-th ordinate with the k-th prime through an exact logarithmic phase quantization.

The central object is not an unmarked real number. It is the pair

$$\mathfrak{z}_k = (n_k, z_k), \tag{1.2}$$

where n_k records the lifted phase and z_k records the associated positive scale. This distinction is structural. The identity

$$\gamma_k \log z_k = \pi n_k \tag{1.3}$$

can be inverted only after n_k is retained.

The construction has three purposes. First, it produces a canonical marked encoding of every positive zero ordinate. Second, it places the logarithm of an ordinary prime inside a single explicitly controlled phase cell. Third, it defines a theoretical lattice in which a candidate pair (n, z) determines a candidate ordinate $\pi n/\log z$ exactly.

Remark 1.1. All statements concerning the definition, exact inversion, asymptotic bounds, counting density, and prime-power detuning are proved rigorously below. Statements concerning an independent analytic rule that would select the correct marked points without first inserting the ordinates are explicitly labelled as questions, heuristics, or conjectures.

2. Classical input

2.1. **Zero and prime clocks.** Let

$$N(T) = \#\{k : 0 < \gamma_k \le T\} \tag{2.1}$$

with the usual midpoint convention at an ordinate when needed. The Riemann–von Mangoldt formula gives

$$N(T) = \frac{T}{2\pi}\log\left(\frac{T}{2\pi}\right) - \frac{T}{2\pi} + O(\log T), \qquad T \to \infty, \tag{2.2}$$

see [6, Chapter 9]. Inverting the leading term yields

$$\gamma_k \sim \frac{2\pi k}{\log k}. \tag{2.3}$$

The prime number theorem and its inverse form give

$$\pi(x) \sim \frac{x}{\log x}, \tag{2.4}$$

$$p_k \sim k \log k. \tag{2.5}$$

The inverse asymptotic and its refinements are classical; see [5, 1].

Lemma 2.1. *As $k \to \infty$,*

$$\frac{p_k}{\gamma_k} \sim \frac{(\log k)^2}{2\pi}, \tag{2.6}$$

$$\frac{\pi p_k}{\gamma_k} \sim \frac{1}{2}(\log k)^2. \tag{2.7}$$

Proof. Divide (2.5) by (2.3). Multiplication by π gives (2.7). □

2.2. **Mangoldt frequencies.** The von Mangoldt function is supported on prime powers:

$$\Lambda(q) = \begin{cases} \log p, & q = p^r, \quad r \geq 1, \\ 0, & \text{otherwise.} \end{cases} \tag{2.8}$$

For $\Re s > 1$,

$$-\frac{\zeta'}{\zeta}(s) = \sum_{q=2}^{\infty} \frac{\Lambda(q)}{q^s}. \tag{2.9}$$

In the Guinand–Weil explicit formula, the arithmetic frequencies occur at $\pm \log q$ with $q = p^r$ and coefficient scale $\Lambda(q)/\sqrt{q}$ for the critical-line normalization; see [4, 7].

3. Canonical marked z-primes

Definition 3.1 (Local marked z-prime)**.** For each $k \geq 1$, define

$$a_k := \frac{\gamma_k \log p_k}{\pi}, \tag{3.1}$$

$$n_k := \lceil a_k \rceil, \tag{3.2}$$

$$\varepsilon_k := n_k - a_k, \qquad 0 \leq \varepsilon_k < 1, \tag{3.3}$$

$$z_k := \exp\left(\frac{\pi n_k}{\gamma_k}\right), \tag{3.4}$$

$$\mathfrak{z}_k := (n_k, z_k). \tag{3.5}$$

The marked system is

$$\mathfrak{Z} := \{\mathfrak{z}_k : k \geq 1\}. \tag{3.6}$$

Remark 3.2. The ceiling convention chooses the first resonant lattice node at or above p_k. A nearest-node convention would reduce the maximal logarithmic error by a factor of two but would permit $z_k < p_k$. The one-sided convention is retained because it gives a canonical upper cell and an exact one-sided comparison with p_k.

Proposition 3.3 (Exact quantization)**.** *For every $k \geq 1$,*

$$\gamma_k \log z_k = \pi n_k, \tag{3.7}$$

$$\mathrm{e}^{\mathrm{i}\gamma_k \log z_k} = (-1)^{n_k}, \tag{3.8}$$

$$z_k = p_k \exp\left(\frac{\pi \varepsilon_k}{\gamma_k}\right). \tag{3.9}$$

Proof. Equation (3.7) follows directly from (3.4). Exponentiation gives (3.8). Finally, (3.3) and (3.1) give

$$\frac{\pi n_k}{\gamma_k} = \log p_k + \frac{\pi \varepsilon_k}{\gamma_k}, \tag{3.10}$$

which exponentiates to (3.9). □

Theorem 3.4 (Exact recovery and marked bijection)**.** *The map*

$$\Phi : \{\gamma_k : k \geq 1\} \longrightarrow \mathfrak{Z}, \qquad \Phi(\gamma_k) = (n_k, z_k), \tag{3.11}$$

is a bijection when the zero ordinates are counted with their indexing multiplicity. Its inverse is

$$\Phi^{-1}(n, z) = \frac{\pi n}{\log z}. \tag{3.12}$$

Proof. Surjectivity holds by the definition of $\mathfrak{Z}$. If $(n_j, z_j) = (n_k, z_k)$, then (3.7) gives

$$\gamma_j = \frac{\pi n_j}{\log z_j} = \frac{\pi n_k}{\log z_k} = \gamma_k. \tag{3.13}$$

Thus the marked data determine the ordinate exactly. With multiplicities retained at the index level, each indexed zero corresponds to one indexed marked pair. □

Warning 3.5. The projection $(n_k, z_k) \mapsto z_k$ can lose information. Distinct ordinates could in principle produce the same positive coordinate with different integer marks. The exact object is therefore the marked pair, not the unmarked real number.

4. The variable logarithmic lattice

Definition 4.1 (Lattice attached to an ordinate)**.** For $\gamma > 0$, define the signed logarithmic resonance lattice

$$\mathcal{L}(\gamma) := \left\{ \frac{\pi n}{\gamma} : n \in \mathbb{Z} \right\} \subset \mathbb{R}, \tag{4.1}$$

and its positive exponential image

$$\mathcal{E}(\gamma) := \left\{ \exp\left(\frac{\pi n}{\gamma}\right) : n \in \mathbb{Z} \right\}. \tag{4.2}$$

The mesh in logarithmic coordinate is

$$h(\gamma) = \frac{\pi}{\gamma}. \tag{4.3}$$

For the k-th zero, $\log z_k$ is the first node of $\mathcal{L}(\gamma_k)$ at or above $\log p_k$.

Proposition 4.2 (One-cell characterization)**.** *The marked z-prime is uniquely characterized by*

$$\log z_k \in \mathcal{L}(\gamma_k), \tag{4.4}$$

$$\log p_k \leq \log z_k < \log p_k + \frac{\pi}{\gamma_k}. \tag{4.5}$$

Equivalently,

$$\frac{\pi(n_k - 1)}{\gamma_k} < \log p_k \leq \frac{\pi n_k}{\gamma_k}. \tag{4.6}$$

Proof. The ceiling relation $n_k - 1 < a_k \leq n_k$ is equivalent to (4.6). The right endpoint is $\log z_k$ by (3.4), and subtracting $\log p_k$ gives (4.5). There is exactly one first lattice node in that half-open cell. □

Remark 4.3. The construction is a family of lattices rather than a single fixed lattice: the mesh π/γ_k varies with k. The marked sequence selects one node from each member of this family. In the $(n, \log z)$ plane, every selected point determines the ordinate through the level curve $\gamma = \pi n / \log z$.

5. Exact and asymptotic bounds

5.1. Sharp one-cell inequalities.

Theorem 5.1 (Exact local bounds)**.** *For every $k \geq 1$,*

$$0 \leq \log \frac{z_k}{p_k} < \frac{\pi}{\gamma_k}, \tag{5.1}$$

$$p_k \leq z_k < p_k \mathrm{e}^{\pi/\gamma_k}, \tag{5.2}$$

$$0 \leq z_k - p_k < p_k \left(\mathrm{e}^{\pi/\gamma_k} - 1\right). \tag{5.3}$$

Moreover,

$$\frac{\pi(n_k - 1)}{\log p_k} < \gamma_k \leq \frac{\pi n_k}{\log p_k}, \tag{5.4}$$

$$\gamma_k = \frac{\pi n_k}{\log z_k}. \tag{5.5}$$

Proof. Equation (5.1) is (4.5). Exponentiation yields (5.2); subtracting p_k gives (5.3). Rearranging (4.6) yields (5.4). Equation (5.5) is the exact inversion formula. □

Corollary 5.2 (Bounds using only n_k and p_k)**.** *For $n_k > 1$,*

$$p_k \leq z_k < p_k^{n_k/(n_k-1)}. \tag{5.6}$$

Proof. The upper bound in (5.4) gives $\log z_k \geq \log p_k$. The lower bound gives

$$\log z_k = \frac{\pi n_k}{\gamma_k} < \frac{n_k}{n_k - 1} \log p_k, \tag{5.7}$$

which proves (5.6). □

5.2. Asymptotic characterization.

Theorem 5.3 (Asymptotic mark, scale, and displacement)**.** *As $k \to \infty$,*

$$\frac{n_k}{k} \longrightarrow 2, \tag{5.8}$$

$$\frac{z_k}{p_k} \longrightarrow 1, \tag{5.9}$$

$$\frac{z_k}{k \log k} \longrightarrow 1, \tag{5.10}$$

$$z_k - p_k = \left(\frac{\varepsilon_k}{2} + o(1)\right) (\log k)^2. \tag{5.11}$$

The $o(1)$ in (5.11) is uniform for $0 \leq \varepsilon_k < 1$.

Proof. From (3.2) and (3.3),

$$n_k = \frac{\gamma_k \log p_k}{\pi} + \varepsilon_k. \tag{5.12}$$

Using (2.3), $\log p_k \sim \log k$, and $0 \leq \varepsilon_k < 1$ gives (5.8).

By (3.9),

$$\log \frac{z_k}{p_k} = \frac{\pi \varepsilon_k}{\gamma_k} = O\left(\frac{\log k}{k}\right), \tag{5.13}$$

which proves (5.9). Combining with $p_k \sim k \log k$ gives (5.10).

Finally, uniformly for $0 \leq \varepsilon_k < 1$,

$$\mathrm{e}^{\pi \varepsilon_k/\gamma_k} - 1 = \frac{\pi \varepsilon_k}{\gamma_k} + O\left(\frac{1}{\gamma_k^2}\right). \tag{5.14}$$

Multiply by p_k and apply Lemma 2.1. Since $p_k/\gamma_k^2 = o((\log k)^2)$, equation (5.11) follows. □

Corollary 5.4 (Maximal asymptotic envelope)**.** *The displacement satisfies*

$$z_k - p_k = O((\log k)^2), \tag{5.15}$$

$$0 \le \limsup_{k\to\infty} \frac{z_k - p_k}{(\log k)^2} \le \frac{1}{2}. \tag{5.16}$$

More precisely,

$$\frac{z_k - p_k}{(\log k)^2} - \frac{\varepsilon_k}{2} \longrightarrow 0. \tag{5.17}$$

Proof. Apply (5.11) and $0 \le \varepsilon_k < 1$. □

Remark 5.5. The constant $1/2$ is the universal asymptotic upper envelope obtained from the one-cell construction. Equality of the limsup with $1/2$ would require values of ε_k approaching 1 along a subsequence. No such distribution statement is assumed here.

Corollary 5.6 (Scaled logarithmic displacement)**.** *One has*

$$\frac{k}{\log k} \log \frac{z_k}{p_k} = \frac{\varepsilon_k}{2} + o(1). \tag{5.18}$$

Consequently,

$$0 \le \limsup_{k\to\infty} \frac{k}{\log k} \log \frac{z_k}{p_k} \le \frac{1}{2}. \tag{5.19}$$

Proof. Use (5.1), the exact identity $\log(z_k/p_k) = \pi\varepsilon_k/\gamma_k$, and (2.3). □

6. Counting density and generalized-prime interpretation

Definition 6.1 (Marked and projected counting functions)**.** Define

$$\Pi_{\mathfrak{Z}}(x) := \#\{k : z_k \le x\}, \tag{6.1}$$

$$\Pi_{\mathfrak{Z}}(x; A) := \#\{k : z_k \le x,\ n_k \in A\}, \qquad A \subseteq \mathbb{N}. \tag{6.2}$$

The first count includes multiplicity in the projected coordinate.

Theorem 6.2 (Prime-density preservation)**.** *The projected counting function satisfies*

$$\Pi_{\mathfrak{Z}}(x) \sim \pi(x) \sim \frac{x}{\log x}. \tag{6.3}$$

Proof. Since $z_k \ge p_k$, one has

$$\Pi_{\mathfrak{Z}}(x) \le \pi(x). \tag{6.4}$$

Fix $\delta > 0$. By (5.9), for all sufficiently large k,

$$z_k \le (1+\delta)p_k. \tag{6.5}$$

Therefore, apart from finitely many indices, $p_k \le x/(1+\delta)$ implies $z_k \le x$, and hence

$$\pi\left(\frac{x}{1+\delta}\right) - O_\delta(1) \le \Pi_{\mathfrak{Z}}(x) \le \pi(x). \tag{6.6}$$

Divide by $x/\log x$, use the prime number theorem, and let $\delta \downarrow 0$. □

Remark 6.3. The proof does not require the sequence (z_k) to be increasing. Counting is performed on the indexed multiset. If one wishes to place the projected coordinates into the standard formalism of Beurling generalized primes, they may be rearranged nondecreasingly and counted with multiplicity; the Euler product in its half-plane of absolute convergence is independent of that rearrangement. The general framework of generalized primes originates in [2]; see also [3].

Definition 6.4 (Projected Beurling-type zeta function)**.** For $\Re s > 1$, define

$$\zeta_{\mathfrak{Z}}^{\mathrm{proj}}(s) := \prod_{k=1}^{\infty} (1 - z_k^{-s})^{-1}. \tag{6.7}$$

Proposition 6.5 (Absolute convergence)**.** *The product* (6.7) *converges absolutely and locally uniformly for* $\Re s > 1$*.*

Proof. Since $z_k \geq p_k$,

$$\sum_{k=1}^{\infty} z_k^{-\sigma} \leq \sum_{k=1}^{\infty} p_k^{-\sigma} < \infty, \qquad \sigma > 1. \tag{6.8}$$

The standard Euler-product convergence criterion applies. □

Warning 6.6. The projected product (6.7) forgets n_k and therefore does not retain the exact inverse map (3.12). The marked transform introduced in Section 8 is the appropriate object for zero recovery.

7. Mangoldt-frequency resonances

7.1. **The matched prime.** The ordinary prime p_k lies below the exact resonant node z_k in the same logarithmic cell. Its phase is therefore determined by the cell remainder.

Theorem 7.1 (Exact matched-prime phase)**.** *For every* $k \geq 1$*,*

$$\gamma_k \log p_k = \pi(n_k - \varepsilon_k), \tag{7.1}$$

$$\mathrm{e}^{\mathrm{i}\gamma_k \log p_k} = (-1)^{n_k} \mathrm{e}^{-\mathrm{i}\pi\varepsilon_k}. \tag{7.2}$$

The angular distance of $\gamma_k \log p_k$ *from the signed lattice* $\pi\mathbb{Z}$ *is*

$$\operatorname{dist}(\gamma_k \log p_k, \pi\mathbb{Z}) = \pi \min\{\varepsilon_k, 1 - \varepsilon_k\}. \tag{7.3}$$

Proof. Equation (7.1) is the definition of ε_k. Exponentiation yields (7.2). The distance formula follows because n_k is an integer and $0 \leq \varepsilon_k < 1$. □

Remark 7.2. The metric closeness $z_k/p_k \to 1$ does not force $\mathrm{e}^{\mathrm{i}\gamma_k \log p_k}$ to approach ± 1. Multiplication by γ_k expands the logarithmic displacement back to the full cell coordinate $\pi\varepsilon_k$. Thus z_k is close to p_k in scale, while the residual phase within the cell can range across an interval of length π.

7.2. **Prime powers.**

Theorem 7.3 (Exact prime-power detuning)**.** *For every* $r \geq 1$*,*

$$\gamma_k \log p_k^r = \pi r(n_k - \varepsilon_k), \tag{7.4}$$

$$\mathrm{e}^{\mathrm{i}\gamma_k \log p_k^r} = (-1)^{r n_k} \mathrm{e}^{-\mathrm{i}\pi r\varepsilon_k}, \tag{7.5}$$

$$\operatorname{dist}(\gamma_k \log p_k^r, \pi\mathbb{Z}) = \pi \|r\varepsilon_k\|_{\mathbb{R}/\mathbb{Z}}, \tag{7.6}$$

where

$$\|x\|_{\mathbb{R}/\mathbb{Z}} := \min_{m \in \mathbb{Z}} |x - m|. \tag{7.7}$$

Proof. Multiply (7.1) by r. The phase identity follows by exponentiation, and the distance identity follows after removing the integer rn_k. □

Corollary 7.4 (Finite resonant-power approximation)**.** *For every real* ε_k *and every integer* $R \geq 1$*, there exists an integer* $1 \leq r \leq R$ *such that*

$$\operatorname{dist}(\gamma_k \log p_k^r, \pi\mathbb{Z}) \leq \frac{\pi}{R+1}. \tag{7.8}$$

Proof. Apply the pigeonhole form of Dirichlet approximation to the $R+1$ points $0, \varepsilon_k, 2\varepsilon_k, \dots, R\varepsilon_k$ modulo one. Two lie within distance $1/(R+1)$, so for some $1 \le r \le R$, $\|r\varepsilon_k\|_{\mathbb{R}/\mathbb{Z}} \le 1/(R+1)$. Apply (7.6). □

Proposition 7.5 (Exact resonance criterion)**.** *For a fixed k and $r \ge 1$, the prime power p_k^r is exactly signed resonant with γ_k if and only if*

$$r\varepsilon_k \in \mathbb{Z}. \tag{7.9}$$

It is exactly positive resonant if and only if

$$r(n_k - \varepsilon_k) \in 2\mathbb{Z}. \tag{7.10}$$

Proof. The first statement follows from (7.6). The second follows from (7.4) and the condition that the lifted phase be an even multiple of π. □

7.3. **Interaction with explicit-formula weights.** At the prime power $q = p_k^r$, the critical-line arithmetic weight is

$$W_{k,r} := \frac{\Lambda(p_k^r)}{\sqrt{p_k^r}} = \frac{\log p_k}{p_k^{r/2}}. \tag{7.11}$$

Proposition 7.6 (Resonance–amplitude tradeoff)**.** *For every $R \ge 1$, there exists $1 \le r \le R$ satisfying* (7.8)*, but its weight obeys*

$$W_{k,r} \le \frac{\log p_k}{\sqrt{p_k}}, \tag{7.12}$$

and for $r \ge 2$,

$$W_{k,r} \le \frac{\log p_k}{p_k}. \tag{7.13}$$

Thus arbitrarily fine phase approximation obtained by increasing r is accompanied by exponential decay in r at fixed p_k.

Proof. The existence of r is Corollary 7.4. The bounds follow directly from (7.11). □

Definition 7.7 (Weighted resonance score)**.** For $R \ge 1$, define

$$\mathcal{R}_k(R) := \max_{1 \le r \le R} \frac{\log p_k}{p_k^{r/2}} \left|\cos(\pi r\varepsilon_k)\right|. \tag{7.14}$$

Remark 7.8. The score (7.14) separates two mechanisms: phase recurrence of $r\varepsilon_k$ modulo one and Mangoldt-weight decay. The first is Diophantine; the second is arithmetic and strongly favors the smallest exponents. This makes the matched prime and the first few powers the natural range for any analytic test based on the explicit formula.

7.4. **General prime-power frequencies near a marked node.**

Proposition 7.9 (Detuning from an arbitrary prime power)**.** *Let $q = p^r$ be any prime power. Define*

$$\Delta_k(q) := \gamma_k \log\left(\frac{q}{z_k}\right). \tag{7.15}$$

Then

$$e^{i\gamma_k \log q} = (-1)^{n_k} e^{i\Delta_k(q)}. \tag{7.16}$$

Moreover, if

$$\left|\log \frac{q}{z_k}\right| \le \frac{\eta}{\gamma_k}, \tag{7.17}$$

then

$$\left|\mathrm{e}^{\mathrm{i}\gamma_k \log q} - (-1)^{n_k}\right| \le \eta. \tag{7.18}$$

Proof. Use $\gamma_k \log z_k = \pi n_k$ to obtain (7.16). Under (7.17), one has $|\Delta_k(q)| \le \eta$, and $|\mathrm{e}^{\mathrm{i}x} - 1| \le |x|$ gives (7.18). □

8. Marked analytic transforms

Definition 8.1 (Marked atomic measure)**.** Define the measure on $\mathbb{N} \times (1, \infty)$

$$\mathrm{d}\Pi_{\mathfrak{Z}}(n, x) := \sum_{k \ge 1} \delta_{n_k}(n)\, \delta_{z_k}(x). \tag{8.1}$$

Definition 8.2 (Marked Dirichlet transform)**.** For $\Re s > 1$ and $|u| \le 1$, define

$$D_{\mathfrak{Z}}(s, u) := \sum_{k \ge 1} u^{n_k} z_k^{-s}, \tag{8.2}$$

$$= \int_{\mathbb{N} \times (1,\infty)} u^n x^{-s}\, \mathrm{d}\Pi_{\mathfrak{Z}}(n, x). \tag{8.3}$$

Proposition 8.3 (Absolute convergence of the marked transform)**.** *The series* (8.2) *converges absolutely and locally uniformly for* $\Re s > 1$ *and* $|u| \le 1$*.*

Proof. Since $|u|^{n_k} \le 1$ and $z_k \ge p_k$,

$$\sum_{k \ge 1} |u|^{n_k} z_k^{-\sigma} \le \sum_{k \ge 1} p_k^{-\sigma} < \infty, \qquad \sigma > 1. \tag{8.4}$$

□

Definition 8.4 (Marked Euler transform)**.** For $\Re s > 1$ and $|u| < 1$, define

$$\zeta_{\mathfrak{Z}}(s, u) := \prod_{k \ge 1} \left(1 - u^{n_k} z_k^{-s}\right)^{-1}. \tag{8.5}$$

Proposition 8.5 (Logarithmic expansion)**.** *For* $\Re s > 1$ *and* $|u| < 1$*,*

$$\log \zeta_{\mathfrak{Z}}(s, u) = \sum_{m \ge 1} \frac{1}{m} \sum_{k \ge 1} u^{m n_k} z_k^{-ms}. \tag{8.6}$$

The double series converges absolutely and locally uniformly.

Proof. Expand $-\log(1 - w) = \sum_{m \ge 1} w^m/m$ with $w = u^{n_k} z_k^{-s}$. Absolute convergence follows from $|u| < 1$, $z_k \ge 2$, and Proposition 8.3. □

Proposition 8.6 (Coefficient decomposition)**.** *For* $|u| < 1$*,*

$$D_{\mathfrak{Z}}(s, u) = \sum_{n \ge 1} A_n(s) u^n, \tag{8.7}$$

where

$$A_n(s) = \sum_{k : n_k = n} z_k^{-s}. \tag{8.8}$$

Thus the coefficient indexed by n retains precisely the projected scales carrying that phase lift.

Proof. Group the absolutely convergent series (8.2) by equal values of n_k. □

9. Theoretical localization of zero ordinates

Definition 9.1 (Inverse lattice map)**.** For $n \in \mathbb{N}$ and $z > 1$, define

$$\Gamma(n, z) := \frac{\pi n}{\log z}. \tag{9.1}$$

Proposition 9.2 (Exact localization on the marked support)**.** *For every atom* (n_k, z_k) *of* (8.1)*,*

$$\Gamma(n_k, z_k) = \gamma_k. \tag{9.2}$$

Proof. This is (5.5). □

Proposition 9.3 (Stability of the inverse coordinate)**.** *Let* $n \geq 1$*,* $z > 1$*, and perturb the scale to* $z\mathrm{e}^{\delta}$*. Then*

$$\Gamma(n, z\mathrm{e}^{\delta}) - \Gamma(n, z) = -\frac{\pi n \delta}{\log z\,(\log z + \delta)}. \tag{9.3}$$

If $|\delta| \leq \frac{1}{2}\log z$*, then*

$$\left|\Gamma(n, z\mathrm{e}^{\delta}) - \Gamma(n, z)\right| \leq \frac{2\pi n}{(\log z)^2}|\delta|. \tag{9.4}$$

Proof. Subtract the two rational expressions $\pi n/(\log z + \delta)$ and $\pi n/\log z$. The stated denominator bound gives (9.4). □

Corollary 9.4 (Stability near the k-th marked point)**.** *At* (n_k, z_k)*, a logarithmic error* δ *produces, to first order,*

$$\Gamma(n_k, z_k\mathrm{e}^{\delta}) - \gamma_k = -\frac{\gamma_k}{\log z_k}\delta + O\left(\frac{\gamma_k}{(\log z_k)^2}\delta^2\right). \tag{9.5}$$

Since $\gamma_k/\log z_k \sim 2\pi k/(\log k)^2$*, localization of* γ_k *to bounded absolute error requires logarithmic scale accuracy of order at most* $(\log k)^2/k$*.*

Proof. Expand (9.3) at $\delta = 0$ and use (2.3) and (5.10). □

Heuristic 9.5 (Marked-transform localization)**.** Suppose an analytic continuation or a regularized inversion procedure recovers the atomic support of $\mathrm{d}\Pi_{\mathfrak{Z}}$ from $D_{\mathfrak{Z}}(s, u)$ or $\zeta_{\mathfrak{Z}}(s, u)$. Coefficient extraction in u separates the integer marks, Mellin inversion in s recovers the corresponding positive scales, and (9.1) then returns the ordinates. This is a heuristic program until an independent construction of the marked transform is available that does not insert the ordinates into (3.2)–(3.4).

Question 9.6. Is there a functional equation, extremal principle, trace identity, or regularized Euler relation whose solutions select the same marked support $\{(n_k, z_k)\}$ from the ordinary prime data and the integer lattice alone?

Question 9.7. Can one define an operator on approximate ordinate sequences by the cycle

$$\{t_k\} \longmapsto \{(n_k(t), z_k(t))\} \longmapsto \text{zeros of a regularized marked transform} \tag{9.6}$$

for which the actual ordinates form a locally attracting fixed point?

10. Comparison with ordinary prime frequencies

Proposition 10.1 (Frequency-cell width)**.** *The difference between the marked logarithmic frequency and the matched prime frequency is*

$$0 \leq \log z_k - \log p_k < \frac{\pi}{\gamma_k} \sim \frac{\log k}{2k}. \tag{10.1}$$

Thus each $\log p_k$ *lies within one vanishing cell below an exact signed resonance for* γ_k*.*

Proof. The exact inequality is (5.1). The asymptotic cell width follows from (2.3). □

Proposition 10.2 (First-order replacement of prime Dirichlet atoms)**.** *Fix a compact set $K \subset \{s \in \mathbb{C} : \Re s > 0\}$. Uniformly for $s \in K$,*

$$z_k^{-s} - p_k^{-s} = -s\frac{\pi\varepsilon_k}{\gamma_k}p_k^{-s} + O_K\left(\frac{p_k^{-\Re s}}{\gamma_k^2}\right). \tag{10.2}$$

Proof. Use (3.9) to write

$$z_k^{-s} = p_k^{-s}\exp\left(-s\frac{\pi\varepsilon_k}{\gamma_k}\right), \tag{10.3}$$

and apply the Taylor expansion of the exponential uniformly on K. □

Corollary 10.3 (Relative prime-series convergence)**.** *For every $\sigma > 0$,*

$$\sum_{k\geq 1} |z_k^{-s} - p_k^{-s}| \tag{10.4}$$

converges locally uniformly in $\Re s \geq \sigma$.

Proof. By (10.2), it is enough to sum $p_k^{-\sigma}/\gamma_k$. The classical asymptotics give

$$\frac{p_k^{-\sigma}}{\gamma_k} \ll_\sigma \frac{(\log k)^{1-\sigma}}{k^{1+\sigma}}, \tag{10.5}$$

which is summable for every $\sigma > 0$. □

Theorem 10.4 (Holomorphic relative Euler factor)**.** *For $\Re s > 1$, set*

$$H(s) := \frac{\zeta_{\mathfrak{Z}}^{\mathrm{proj}}(s)}{\zeta(s)}. \tag{10.6}$$

Then H extends to a holomorphic, nonvanishing function on $\Re s > 0$ through

$$\log H(s) = \sum_{m\geq 1}\frac{1}{m}\sum_{k\geq 1}\left(z_k^{-ms} - p_k^{-ms}\right). \tag{10.7}$$

Consequently, wherever ζ is meromorphic in $\Re s > 0$,

$$\zeta_{\mathfrak{Z}}^{\mathrm{proj}}(s) = H(s)\zeta(s) \tag{10.8}$$

provides the corresponding meromorphic continuation of the projected z-prime Euler product.

Proof. For $m = 1$, local uniform convergence on every half-plane $\Re s \geq \sigma > 0$ follows from Corollary 10.3. For $m \geq 2$,

$$|z_k^{-ms} - p_k^{-ms}| \leq z_k^{-m\sigma} + p_k^{-m\sigma} \leq 2p_k^{-m\sigma}. \tag{10.9}$$

For fixed $\sigma > 0$, finitely many m with $m\sigma \leq 1$ are handled by the same difference estimate, and the remaining tail converges absolutely by comparison with a geometric series in $p_k^{-\sigma}$. Thus (10.7) is holomorphic on $\Re s > 0$. Exponentiation gives a holomorphic nonvanishing H. On $\Re s > 1$, the Euler-product logarithms show that it equals the quotient (10.6); analytic continuation gives (10.8). □

Remark 10.5. The factorization (10.8) is a rigorous analytic consequence of the small displacement. It shows that the projected Euler system differs from the ordinary prime system by a regular nonvanishing factor in $\Re s > 0$. The marked transform contains additional information not present in this projection.

11. Index geometry and finite candidate windows

Proposition 11.1 (Asymptotic diagonal)**.** *The marked points lie asymptotically on the diagonal $n = 2k$:*

$$n_k = 2k + o(k). \tag{11.1}$$

Moreover,

$$\log z_k = \log k + \log\log k + o(1). \tag{11.2}$$

Proof. Equation (11.1) is equivalent to (5.8). Equation (11.2) follows from $z_k \sim k \log k$. □

Heuristic 11.2 (Finite search strip)**.** For a large index k, the asymptotic relation $n_k = 2k + o(k)$ restricts the integer mark to a narrow relative strip around $2k$, while $z_k = p_k + O((\log k)^2)$ restricts the scale to a polylogarithmic interval above p_k. A theoretical search may therefore be organized over candidate pairs

$$n \in [2k - E_k, 2k + E_k] \cap \mathbb{Z}, \qquad z \in [p_k, p_k + C(\log k)^2], \tag{11.3}$$

with $E_k = o(k)$ and any fixed $C > 1/2$ for sufficiently large k. An independent selector is still required to choose the actual pair inside this strip.

12. Alternative conventions

12.1. **Nearest-node convention.**

Definition 12.1 (Nearest marked z-prime)**.** Define

$$n_k^{\mathrm{near}} := \left\lfloor \frac{\gamma_k \log p_k}{\pi} + \frac{1}{2} \right\rfloor, \qquad z_k^{\mathrm{near}} := \exp\left(\frac{\pi n_k^{\mathrm{near}}}{\gamma_k} \right). \tag{12.1}$$

Proposition 12.2 (Nearest-node error)**.** *The nearest-node convention satisfies*

$$\left| \log \frac{z_k^{\mathrm{near}}}{p_k} \right| \le \frac{\pi}{2\gamma_k} \tag{12.2}$$

and therefore

$$|z_k^{\mathrm{near}} - p_k| \le \left(\frac{1}{4} + o(1) \right) (\log k)^2. \tag{12.3}$$

Proof. Nearest-integer rounding gives an error at most $1/2$ in the lifted cell coordinate. The additive asymptotic follows as in Theorem 5.3. □

Remark 12.3. The nearest convention minimizes metric displacement, while the ceiling convention preserves $z_k \ge p_k$ and gives a canonical upper node. Both retain exact recovery when the integer mark is stored. The one-sided convention is preferable when comparing counting functions by monotone inequalities; the nearest convention is preferable when minimizing local perturbation.

12.2. **Positive-phase convention.**

Definition 12.4 (Positive-phase lattice)**.** Define

$$n_k^{+} := \left\lceil \frac{\gamma_k \log p_k}{2\pi} \right\rceil, \qquad z_k^{+} := \exp\left(\frac{2\pi n_k^{+}}{\gamma_k} \right). \tag{12.4}$$

Proposition 12.5 (Positive-phase cost)**.** *One has*

$$\mathrm{e}^{\mathrm{i}\gamma_k \log z_k^{+}} = 1, \tag{12.5}$$

$$0 \le \log \frac{z_k^{+}}{p_k} < \frac{2\pi}{\gamma_k}, \tag{12.6}$$

$$z_k^{+} - p_k \le (1 + o(1))(\log k)^2. \tag{12.7}$$

Proof. The proof repeats the ceiling argument with lattice mesh $2\pi/\gamma_k$. Lemma 2.1 gives the final envelope. □

Remark 12.6. The signed lattice has half the mesh and therefore half the maximal one-sided displacement. Retaining the parity of n_k records whether the exact node has phase $+1$ or -1.

13. Open analytic problems

Question 13.1. What is the distribution of

$$\varepsilon_k = \left\lceil \frac{\gamma_k \log p_k}{\pi} \right\rceil - \frac{\gamma_k \log p_k}{\pi} \tag{13.1}$$

in $[0,1)$? Any equidistribution or discrepancy theorem would immediately refine the displacement statistics and the prime-power resonance counts.

Conjecture 13.2 (Cell equidistribution)**.** *The sequence* (ε_k) *is equidistributed in* $[0,1)$.

Remark 13.3. Conjecture 13.2 would imply, for every continuous f on $[0,1]$,

$$\frac{1}{K} \sum_{k \le K} f\left(\frac{2(z_k - p_k)}{(\log k)^2} \right) \longrightarrow \int_0^1 f(t)\, \mathrm{d}t, \tag{13.2}$$

provided the uniform error in (5.11) is incorporated. It would also give statistical information on the resonance factors $\cos(\pi r \varepsilon_k)$.

Question 13.4. Can the marked measure (8.1) be realized as the spectral measure of an operator or as the divisor of a two-variable analytic function whose arithmetic side is expressed directly through ordinary primes and prime powers?

Question 13.5. Does there exist a regularization of $D_{\mathfrak{Z}}(s,u)$ at or beyond its natural half-plane that admits stable coefficient extraction in u and Mellin inversion in z, with quantitative control compatible with (9.4)?

Question 13.6. For fixed small R, can one prove nonrandom structure in

$$\min_{1 \le r \le R} \| r \varepsilon_k \|_{\mathbb{R}/\mathbb{Z}} \tag{13.3}$$

that correlates with the low prime-power modes in a regularized explicit formula?

14. Summary of proved characterizations

The complete marked construction is

$$\boxed{n_k = \left\lceil \frac{\gamma_k \log p_k}{\pi} \right\rceil, \qquad z_k = \exp\left(\frac{\pi n_k}{\gamma_k} \right), \qquad \mathfrak{z}_k = (n_k, z_k).} \tag{14.1}$$

Its exact inverse is

$$\boxed{\gamma_k = \frac{\pi n_k}{\log z_k}.} \tag{14.2}$$

The one-cell bounds are

$$\boxed{0 \le \log \frac{z_k}{p_k} < \frac{\pi}{\gamma_k}, \qquad 0 \le z_k - p_k < p_k \left(\mathrm{e}^{\pi/\gamma_k} - 1 \right).} \tag{14.3}$$

The principal asymptotics are

$$\boxed{n_k \sim 2k, \qquad z_k \sim p_k \sim k \log k, \qquad z_k - p_k = \left(\frac{\varepsilon_k}{2} + o(1) \right) (\log k)^2.} \tag{14.4}$$

The counting law is

$$\boxed{\Pi_{\mathfrak{Z}}(x) \sim \pi(x) \sim \frac{x}{\log x}.} \tag{14.5}$$

The exact matched prime-power resonance is

$$\boxed{\mathrm{e}^{\mathrm{i}\gamma_k \log p_k^r} = (-1)^{rn_k}\mathrm{e}^{-\mathrm{i}\pi r\varepsilon_k}, \qquad \operatorname{dist}(\gamma_k \log p_k^r, \pi\mathbb{Z}) = \pi\|r\varepsilon_k\|_{\mathbb{R}/\mathbb{Z}}.} \tag{14.6}$$

Finally, the theoretical localization coordinate is

$$\boxed{(n, z) \longmapsto \Gamma(n, z) = \frac{\pi n}{\log z}.} \tag{14.7}$$

The marked pair is therefore an exact coordinate for an ordinate, while the analytic problem is to characterize or recover the correct marked support by an independent global relation.

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